

The overlap operator classifies power partial isometric operator-valued Toeplitz operators

Candidate full answer to Question IV in arXiv:2407.10609

13 August 2026

Status: candidate full classification; likely valid. Let T_Φ be an operator-valued Toeplitz operator on $H^2_\mathcal{E}(\mathbb{D}^n)$. The source theorem writes every nonzero partial isometry as $M_\Gamma M_\Psi^*$ with inner Γ, Ψ . Put $C = M_\Psi^* M_\Gamma = T_{\Psi^* \Gamma}$. Then

$$T_\Phi^m = M_\Gamma C^{m-1} M_\Psi^* \quad (m \geq 1),$$

and consequently T_Φ is a power partial isometry if and only if C is. The Halmos–Wallen theorem turns this into an explicit canonical classification of C . Constant symbols are included separately and the zero operator is harmless.

1 The question

For a Hilbert space $\mathcal{E}$, let $H^2_\mathcal{E}(\mathbb{D}^n)$ be the vector-valued Hardy space on the polydisc. Sarkar [1] ends with: “Characterize Toeplitz operators, which are power partial isometries.” Here T is a *power partial isometry* (PPI) when T^m is a partial isometry for every integer $m \geq 1$.

The key input from the source is its classification of partially isometric Toeplitz operators. Apart from the trivial zero operator, such an operator has a factorization

$$T_\Phi = M_\Gamma M_\Psi^*, \tag{1}$$

where $\Gamma, \Psi \in H^\infty_{\mathcal{B}(\mathcal{F}, \mathcal{E})}(\mathbb{D}^n)$ are inner and obey the source’s necessary and sufficient Toeplitz-product condition

$$(\Gamma(\lambda) - \Gamma_k(\lambda))(\Psi(\mu) - \Psi_k(\mu))^* = 0 \tag{2}$$

for every $\lambda, \mu \in \mathbb{D}^n$ and $k = 1, \dots, n$; here the subscript k means that the k th variable is set equal to zero. Equivalently, the corresponding positive Fourier coefficients in the same variable have orthogonal products.

2 Full overlap classification

Write S for the unilateral shift, J_q for the nilpotent truncated shift of length q , and allow arbitrary Hilbert-space multiplicities in tensor products below.

Theorem 1 (Classification). *Let $0 \neq T_\Phi$ be a Toeplitz operator on $H^2_\mathcal{E}(\mathbb{D}^n)$. Then T_Φ is a power partial isometry if and only if there are a Hilbert space $\mathcal{F}$ and inner functions $\Gamma, \Psi \in H^\infty_{\mathcal{B}(\mathcal{F}, \mathcal{E})}(\mathbb{D}^n)$ such that:*

1. $T_\Phi = M_\Gamma M_\Psi^*$ and (2) holds;

8. SOME RELATED QUESTIONS

Based on the results of this article, we can ask several interesting questions for Toeplitz operators on vector-valued Hardy spaces worthy of further investigation.

(I) Characterize Toeplitz operators with shift-invariant range spaces.

A significant contribution of this article is to show that if we start with a somewhat nice Toeplitz operator, then the range is Beurling-type. This gives rise to a fascinating problem.

(II) Characterize Toeplitz operators with Beurling-type range spaces.

In Corollary 5.4, we have seen that a partially isometric Toeplitz operator has a partially isometric symbol, a.e. on $\mathbb{T}^n$, but what can we say about the converse?

(III) Characterize symbols $\Phi \in L^\infty_{\mathcal{B}(\mathcal{E})}(\mathbb{T}^n)$ such that T_Φ is a partial isometry.

Unlike in the scalar cases, it is still not clear when partially isometric Toeplitz operators on $H^2_{\mathcal{E}}(\mathbb{D}^n)$ become power partial isometries, so we end with the following question.

(IV) Characterize Toeplitz operators, which are power partial isometries.

ACKNOWLEDGEMENT

The author acknowledges the hospitality extended by the Indian Institute of Science, where

Figure 1: The complete closing question block in arXiv:2407.10609, current PDF page 35.

2. the overlap Toeplitz operator

$$C := M_\Psi^* M_\Gamma = T_{\Psi^* \Gamma} \quad \text{on } H^2_{\mathcal{F}}(\mathbb{D}^n) \quad (3)$$

is unitarily equivalent to

$$U \oplus (S \otimes I_{\mathcal{K}_s}) \oplus (S^* \otimes I_{\mathcal{K}_b}) \oplus \bigoplus_{q \geq 1} (J_q \otimes I_{\mathcal{K}_q}), \quad (4)$$

where U is unitary and any of the multiplicity spaces may be zero.

The zero Toeplitz operator is a PPI. If $T_\Phi = I_{H^2(\mathbb{D}^n)} \otimes A$ is constant, the criterion reduces exactly to: A has the decomposition (4), or equivalently A is a PPI.

The criterion is independent of the chosen inner factorization: for every factorization (1), the associated overlap is a PPI exactly when T_Φ is. Thus nonuniqueness of Γ, Ψ creates no ambiguity.

Lemma 2 (Isometric outer factors reflect partial isometries). *Let $V, W : \mathcal{K} \rightarrow \mathcal{H}$ be isometries and $A \in \mathcal{B}(\mathcal{K})$. Then VAW^* is a partial isometry if and only if A is a partial isometry.*

Proof. One has

$$(VAW^*)^*(VAW^*) = W(A^*A)W^*.$$

If A^*A is a projection, so is the right side. Conversely, if the right side is a projection, compressing by W^* and W shows $(A^*A)^2 = A^*A$. Since $A^*A \geq 0$, this is equivalent to A being a partial isometry. $\square$

Lemma 3 (Power identity). *For $T = M_\Gamma M_\Psi^*$ and $C = M_\Psi^* M_\Gamma$,*

$$T^m = M_\Gamma C^{m-1} M_\Psi^* \quad (m \geq 1). \quad (5)$$

Consequently, T^m is a partial isometry if and only if C^{m-1} is.

Proof. The formula is immediate for $m = 1$ and induction inserts the middle product $M_\Psi^* M_\Gamma = C$. Both M_Γ and M_Ψ are isometries because Γ and Ψ are inner. Lemma 2 applies to (5). $\square$

Proof of Theorem 1. Suppose first that T_Φ is a nonzero PPI. It is in particular a partial isometry, so the source factorization theorem gives (1) and (2). Lemma 3, applied for all $m \geq 2$, says that every positive power of C is a partial isometry. The Halmos–Wallen decomposition theorem [2] gives precisely (4).

Conversely, (2) makes $M_\Gamma M_\Psi^*$ Toeplitz, while innerness makes it a partial isometry. Every summand in (4) is a PPI, hence so is C . Lemma 3 now makes every power of T_Φ a partial isometry.

It remains only to justify the stated edge cases. The zero operator is immediate. A constant symbol gives $T_\Phi = I \otimes A$ and $T_\Phi^m = I \otimes A^m$, so partial-isometry status is preserved and reflected term by term. Alternatively, if A is a partial isometry, take $\mathcal{F} = (\ker A)^\perp$, let $\Psi : \mathcal{F} \hookrightarrow \mathcal{E}$ be inclusion, and let $\Gamma = A|_{\mathcal{F}}$. These are constant isometries and $A = \Gamma\Psi^*$, so the overlap proof also covers constants. $\square$

Remark 4 (Why the overlap is again Toeplitz). For $f, g \in H_{\mathcal{F}}^2(\mathbb{D}^n)$,

$$\langle M_\Psi^* M_\Gamma f, g \rangle = \int_{\mathbb{T}^n} \langle \Psi(\zeta)^* \Gamma(\zeta) f(\zeta), g(\zeta) \rangle d\zeta.$$

Thus $C = T_{\Psi^* \Gamma}$, with no extra product hypothesis. The classification therefore stays entirely inside the Toeplitz category.

3 A direct analytic-symbol criterion

The source also proves that an analytic multiplier M_G is a partial isometry exactly when

$$G(\zeta)^* G(\zeta) = P \quad \text{a.e. on } \mathbb{T}^n \tag{6}$$

for one constant orthogonal projection P on the coefficient space. Indeed, this is the boundary form of the source factorization $G = \Theta V^*$ with Θ inner and V a constant isometry.

Corollary 5 (Analytic symbols). For $\Phi \in H_{B(\mathcal{E})}^\infty(\mathbb{D}^n)$, the analytic Toeplitz operator M_Φ is a power partial isometry if and only if, for every $m \geq 1$, there is a constant orthogonal projection P_m such that

$$\Phi(\zeta)^{*m} \Phi(\zeta)^m = P_m \quad \text{for a.e. } \zeta \in \mathbb{T}^n. \tag{7}$$

Proof. Since multiplication respects pointwise products, $M_\Phi^m = M_{\Phi^m}$. Apply (6) to $G = \Phi^m$ for every m . $\square$

This is a symbol-only criterion. It also shows why merely asking for $\Phi(\zeta)$ to be a partial isometry a.e. is insufficient: the initial projection in every power must be independent of the boundary point, and the requirement must persist through all powers.

4 Partial isometry does not imply PPI

The operator-valued case differs sharply from the scalar case.

Proposition 6 (A 2×2 nonconstant counterexample). *Let*

$$A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad \Phi(z) = z_1 A \quad (z \in \mathbb{D}^n).$$

Then $\Phi(\zeta)$ is a partial isometry a.e., and M_Φ is a nonconstant partially isometric Toeplitz operator, but M_Φ^2 is not a partial isometry.

Proof. Put $u = e_1$ and $v = (e_1 + e_2)/\sqrt{2}$. Then $A = uv^*$, $A^*A = vv^*$, and $AA^* = uu^*$, so A is a partial isometry. But

$$A^2 = \langle v, u \rangle A = \frac{1}{\sqrt{2}} A, \quad (A^2)^* A^2 = \frac{1}{2} vv^*,$$

and the last operator is not a projection. Under $H_{\mathbb{C}^2}^2(\mathbb{D}^n) = H^2(\mathbb{D}^n) \otimes \mathbb{C}^2$, $M_\Phi = M_{z_1} \otimes A$. Hence it is a partial isometry, whereas its square is $M_{z_1^2} \otimes A^2$ and is not. $\square$

More generally, $z_1 A$ is a PPI exactly when the coefficient partial isometry A is a PPI. This embeds every Halmos–Wallen possibility, and every obstruction, into a nonconstant analytic Toeplitz symbol.

5 Focused proof and upgrade audits

Eight focused checks were completed before promotion:

1. The current 36-page arXiv PDF was checked; Question IV remains verbatim on page 35.
2. The source factorization was checked with domains and codomains written explicitly, including a direct treatment of constant and zero symbols.
3. Formula (5) was checked inductively for all powers, rather than inferred from the first few cases.
4. Lemma 2 was checked in both directions; no surjectivity of the outer isometries is used.
5. The overlap identity $C = T_{\Psi^* \Gamma}$ was checked directly through Hardy-space inner products.
6. The Halmos–Wallen theorem was applied only after the overlap was proved to be a PPI; all unitary, shift, co-shift, and truncated-shift summands and multiplicities are retained.
7. The analytic criterion was independently derived by applying the source’s analytic partial-isometry theorem to every pointwise power.
8. The 2×2 counterexample was verified exactly and by a small symbolic check; its square has nonzero singular value $1/\sqrt{2}$ and hence is not a partial isometry.

6 Novelty and review recommendation

A bounded search on 13 August 2026 covered the exact wording of Question IV, the source title and citations, overlap/compressed-inner-product formulations, the scalar-valued classification [3], the 2025 papers of Babbar–Maji on PPIs and characteristic functions [4, 5], and arXiv records through 2026. The 2025 characteristic-function paper advances Question III in one variable but does not

state the overlap reduction or answer Question IV. No later occurrence of this classification was found. Novelty is plausible, not certified, and priority is not claimed.

The principal human-review point is conceptual: decide whether the overlap canonical form is accepted as the requested “characterization.” It is a necessary-and-sufficient parametrization with no unresolved mathematical condition, because (4) is the complete Halmos–Wallen canonical form. The core new proof itself is the two-line power identity plus the fact that isometric outer factors preserve and reflect partial-isometry status.

References

- [1] S. Sarkar, *On products and partial isometry of Toeplitz operators with operator-valued symbols*, arXiv:2407.10609, current PDF 36 pages.
- [2] P. R. Halmos and L. J. Wallen, *Powers of partial isometries*, Indiana Univ. Math. J. **19** (1970), 657–663.
- [3] K. D. Deepak, D. Pradhan, and J. Sarkar, *Partially isometric Toeplitz operators on the polydisc*, Bull. Lond. Math. Soc. **54** (2022), 1350–1362.
- [4] K. Babbar and A. Maji, *Power partial isometries*, arXiv:2503.22707.
- [5] K. Babbar and A. Maji, *Characteristic function of a power partial isometry*, arXiv:2505.07824.